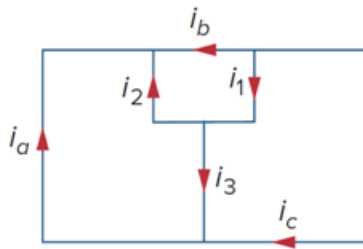


## Problem

Design a problem, complete with a solution, to help other students better understand Kirchhoff's Current Law. Design the problem by specifying values of  $i_a$ ,  $i_b$ , and  $i_c$ , shown in Fig. 2.72, and asking them to solve for values of  $i_1$ ,  $i_2$ , and  $i_3$ . Be careful to specify realistic currents.

**Figure 2.72:**



## Step-by-step solution

## Step 1 of 4

Refer to circuit diagram in Figure 2.72 in the textbook.

Draw the circuit with node notations.

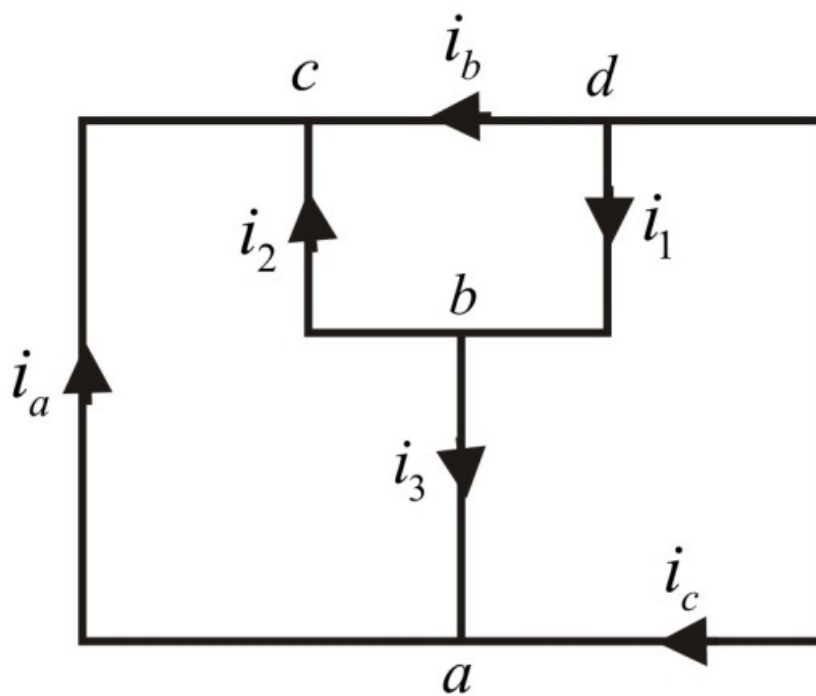


Figure 1

## Step 2 of 4

Let the current values be,

$$i_a = -15 \text{ A}$$

$$i_b = 10 \text{ A}$$

$$i_c = 5 \text{ A}$$

Apply Kirchhoff's current law at node a.

$$i_3 + i_c - i_a = 0$$

$$\begin{aligned} i_3 &= i_a - i_c \\ &= -15 - 5 \text{ A} \\ &= -20 \text{ A} \end{aligned}$$

### Step 3 of 4

Apply Kirchhoff's current law at node c.

$$i_b + i_2 + i_a = 0$$

$$\begin{aligned} i_2 &= -(i_a + i_b) \\ &= -(-15 + 10) \\ &= 5 \text{ A} \end{aligned}$$

Apply Kirchhoff's current law at node d.

$$-(i_b + i_1 + i_c) = 0$$

$$\begin{aligned} i_1 &= -(i_b + i_c) \\ &= -(10 + 5) \\ &= -15 \text{ A} \end{aligned}$$

Hence, the currents are,

$$\boxed{\begin{array}{l} i_1 = -15 \text{ A} \\ i_2 = 5 \text{ A} \\ i_3 = -20 \text{ A} \end{array}}.$$

### Step 4 of 4

Now apply Kirchhoff's current law at node b.

$$\begin{aligned} i_1 - i_2 - i_3 &= -15 - 5 - (-20) \\ &= -20 + 20 \\ &= 0 \end{aligned}$$

Hence the solution is verified.